



آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

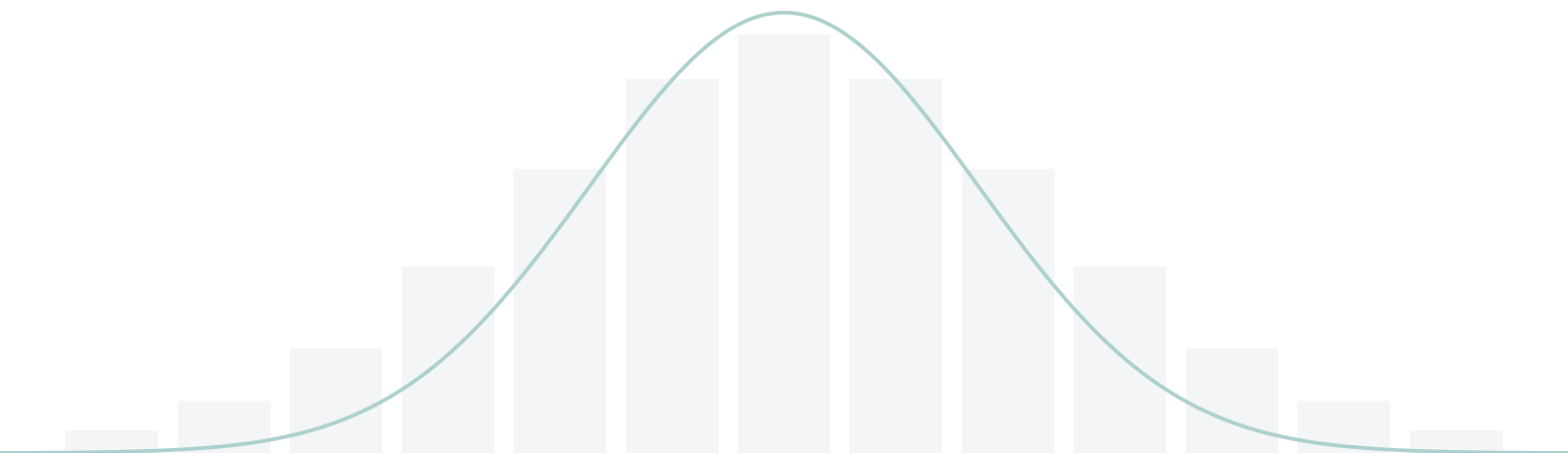
Module 14

Confidence Intervals

بازه‌های اطمینان

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Learning Objectives

After completing this module, students will be able to:

1. Correctly interpret confidence intervals (frequentist interpretation)
2. Construct CIs for means (known and unknown variance)
3. Apply the t-distribution for small samples
4. Construct CIs for variance
5. Determine required sample sizes
6. Avoid common misconceptions about CIs

14.1 Introduction: Why Point Estimates Aren't Enough

A point estimate (e.g., $\bar{X} = 3.47$) gives no information about **precision**. We need to quantify uncertainty:

“The true mean is estimated to be 3.47, **give or take** some amount.”

A **confidence interval** provides a range that, with a specified confidence level, captures the true parameter.

14.2 Correct Interpretation of Confidence Intervals

■ ✓ CORRECT (Frequentist) Interpretation

“If we repeat this experiment many times, each time computing a 95% CI, then approximately 95% of those intervals will contain the true parameter μ .”

The **procedure** has a 95% success rate. Each specific interval either contains μ or it doesn't.

■ ✗ WRONG Interpretation

“There is a 95% probability that μ is inside this particular interval [2.1, 4.8].”

Why this is wrong: μ is a fixed (non-random) number. It's either in [2.1, 4.8] or not — there is no probability about it. The **interval** is random (varies from sample to sample), not the parameter.

■ Analogy

Think of CIs like a fishing net:

- ▶ Before you cast: 95% chance your net will catch the fish (the procedure works 95% of the time)
- ▶ After you cast: the fish is either caught or not — no probability anymore
- ▶ The fish (μ) doesn't move; your net (CI) does

14.3 CI for Mean — Known Variance (Z-interval)

■ Setup

Data: $X_1, \dots, X_n$ i.i.d. from $N(\mu, \sigma^2)$ with σ^2 **known**.

■ Derivation

$\bar{X} \sim N(\mu, \sigma^2/n)$, so $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0,1)$

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$$

Solving for μ :

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

■ Formula

$$\text{CI: } \bar{X} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

■ Common z-values

Confidence Level	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.960
99%	0.01	2.576

Example

Noise measurements: $n = 36$, $\bar{X} = 0.52\text{V}$, $\sigma = 0.12\text{V}$ (known from equipment spec).

95% CI: $0.52 \pm 1.96 \times 0.12/\sqrt{36} = 0.52 \pm 0.0392 = [0.481, 0.559] \text{ V}$

14.4 CI for Mean — Unknown Variance (t-interval)

■ The Problem

Usually σ is unknown. We estimate it with S . But $\bar{X}/(S/\sqrt{n})$ does NOT follow $N(0,1)$.

■ Student's t-Distribution

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

The t-distribution:

- ▶ Symmetric about 0
- ▶ Heavier tails than $N(0,1)$ (more uncertainty because S is estimated)
- ▶ Degrees of freedom: $\nu = n - 1$
- ▶ As $\nu \rightarrow \infty$: $t(\nu) \rightarrow N(0,1)$
- ▶ For $\nu \geq 30$: $t \approx z$ (practically)

■ Formula

$$\text{CI: } \bar{X} \pm t_{\alpha/2, n-1} \cdot \frac{S}{\sqrt{n}}$$

Example

Component delay: $n = 15$, $\bar{X} = 2.34$ ns, $S = 0.45$ ns.

95% CI: $t_{\{0.025, 14\}} = 2.145$

CI: $2.34 \pm 2.145 \times 0.45/\sqrt{15} = 2.34 \pm 0.249 = [2.09, 2.59]$ ns

(Wider than z-interval because of additional uncertainty in S .)

14.5 CI for Variance

■ Setup

Data from $N(\mu, \sigma^2)$. Want CI for σ^2 .

■ Key Distribution

$$(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$$

■ Formula

$$\left[\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2} \right]$$

Example

Noise power: $n = 20$, $S^2 = 0.045 \text{ V}^2$.

95% CI for σ^2 : $\chi^2_{\{0.025, 19\}} = 32.852$, $\chi^2_{\{0.975, 19\}} = 8.907$

CI: $[(19)(0.045)/32.852, (19)(0.045)/8.907] = [0.026, 0.096] \text{ V}^2$

95% CI for σ : $[\sqrt{0.026}, \sqrt{0.096}] = [0.161, 0.310] \text{ V}$

14.6 CI for Proportions

■ Setup

n trials, $\hat{x} = k/n$ successes. Estimate $p = \text{true proportion}$.

■ Large-Sample Formula (Wald Interval)

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Example: Bit Error Rate

Transmitted 100,000 bits, observed 53 errors. $\hat{p} = 53/100000 = 5.3 \times 10^{-4}$.

95% CI: $5.3 \times 10^{-4} \pm 1.96 \sqrt{(5.3 \times 10^{-4} \times 0.9995 / 100000)} = 5.3 \times 10^{-4} \pm 1.43 \times 10^{-4}$

CI: **$[3.9 \times 10^{-4}, 6.7 \times 10^{-4}]$**

14.7 Factors Affecting CI Width

Factor	Effect on Width	Explanation
↑ Sample size n	Narrower	More data → more precision
↑ Confidence level	Wider	More confidence → wider net
↑ Variability σ	Wider	More noise → harder to pin down μ

■ The Tradeoff

- ▶ Want narrow CI (precise) → need more data or accept lower confidence
- ▶ Want high confidence → accept wider CI or collect more data
- ▶ Want both narrow AND high confidence → need LOTS of data

14.8 Sample Size Determination

■ Question: How many measurements do I need?

For CI width $\leq 2E$ (margin of error E):

$$n \geq \left(\frac{z_{\alpha/2} \cdot \sigma}{E} \right)^2$$

Example

Want to estimate mean delay within $\pm 0.5\text{ms}$ with 95% confidence. Expect $\sigma \approx 3\text{ms}$.

$n \geq (1.96 \times 3 / 0.5)^2 = (11.76)^2 = 138.3 \rightarrow \text{need } \mathbf{n = 139 \text{ measurements}}$

14.9 MATLAB Demonstrations

Example 1: CI Coverage Demonstration

MATLAB

```
%% Demonstrate: ~95% of 95% CIs contain the true mean
mu_true = 5; sigma = 2; n = 25;
N_experiments = 1000;
alpha = 0.05;
z_crit = norminv(1-alpha/2);

covers = 0;
figure; hold on;
for i = 1:N_experiments
    data = mu_true + sigma*randn(1, n);
    x_bar = mean(data);
    margin = z_crit * sigma / sqrt(n);
    ci_low = x_bar - margin;
    ci_high = x_bar + margin;

    if ci_low <= mu_true && mu_true <= ci_high
        covers = covers + 1;
        color = 'b';
    else
        color = 'r';
    end

    if i <= 100 % Plot first 100
        plot([ci_low ci_high], [i i], color, 'LineWidth', 0.5);
    end
end
plot([mu_true mu_true], [0 101], 'k--', 'LineWidth', 2);
xlabel('μ'); ylabel('Experiment #');
title(sprintf('95% CI Coverage: %.1f%% of %d intervals contain μ', ...
    100*covers/N_experiments, N_experiments));
fprintf('Coverage: %.1f%% (theory: 95%)\n', 100*covers/N_experiments);
```

Example 2: t-interval in Practice

```

%% t-Confidence Interval: Unknown Variance
data = [2.3, 2.7, 2.1, 2.8, 2.5, 2.9, 2.4, 2.6, 2.2, 2.7];
n = length(data);
x_bar = mean(data);
s = std(data);
alpha = 0.05;

t_crit = tinv(1-alpha/2, n-1);
margin = t_crit * s / sqrt(n);

fprintf('Sample mean: %.3f\n', x_bar);
fprintf('Sample std: %.3f\n', s);
fprintf('t_crit (df=%d): %.3f\n', n-1, t_crit);
fprintf('95% CI: [%.3f, %.3f]\n', x_bar-margin, x_bar+margin);

```

MATLAB

Example 3: Sample Size Calculation

```

%% How many samples do we need?
sigma = 0.1;           % Expected std dev (from pilot study)
E_values = [0.05, 0.02, 0.01, 0.005]; % Desired margins
confidence = 0.95;
z = norminv(1 - (1-confidence)/2);

fprintf('Desired Margin | Required n\n');
fprintf('-----+-----\n');
for E = E_values
    n_req = ceil((z*sigma/E)^2);
    fprintf(' ±%.3f      |    %d\n', E, n_req);
end

```

MATLAB

14.10 Practice Problems**Problem 1**

A sensor measures voltage. From 50 measurements: $\bar{X} = 3.32\text{V}$, $S = 0.15\text{V}$.

- (a) Construct 95% CI for μ . (b) Construct 99% CI for μ . (c) Which is wider and why?

Solution

(a) $t_{0.025,49} \approx 2.010$. CI: $3.32 \pm 2.010(0.15/\sqrt{50}) = 3.32 \pm 0.043 = [3.277, 3.363]$

(b) $t_{0.005,49} \approx 2.680$. CI: $3.32 \pm 2.680(0.15/\sqrt{50}) = 3.32 \pm 0.057 = [3.263, 3.377]$

(c) 99% CI is wider — higher confidence requires a wider interval.

Problem 2

From 25 noise power measurements: $S^2 = 0.048 \text{ V}^2$. Construct 95% CI for σ^2 .

Solution $\chi^2_{0.025,24} = 39.364$, $\chi^2_{0.975,24} = 12.401$

CI: $[24(0.048)/39.364, 24(0.048)/12.401] = [0.029, 0.093] \text{ V}^2$

Problem 3

How many BER measurements needed to estimate BER within $\pm 20\%$ of its true value with 95% confidence, if expected BER $\approx 10^{-3}$?

Solution For proportion: $n \geq z^2 p(1-p)/E^2$ where $E = 0.2p = 2 \times 10^{-4}$.

$n \geq (1.96)^2(10^{-3})(0.999)/(2 \times 10^{-4})^2 = 3.84 \times 10^{-3} / (4 \times 10^{-8}) \approx 96,000$ bits.

Need to transmit approximately 96,000 bits (or observe ~ 96 errors).

Next Module: Module 15 — Hypothesis Testing